

Clinical Features and Classification of Langerhans Cell Histiocytosis (LCH)

All patients with Langerhans cell histiocytosis (LCH) fall along a clinical spectrum of disease, ranging from localized, self-limited forms to life-threatening multisystem involvement.

Cutaneous Manifestations

Cutaneous lesions vary depending on the clinical subtype:

- **Hashimoto-Pritzker disease (HPD):** Traditionally characterized by the eruption of multiple or solitary elevated, firm, red-brown nodules or flesh-red lesions resembling infantile angiomas. Late papulonodular lesions may have elevated borders and are prone to ulceration. Lesions may increase in size and number during the first few weeks of life, sometimes becoming large before spontaneously resolving. They subsequently form brown crusts, which shed and may leave whitish atrophic scars.

Notably, small reddish-brown crusted papules—resembling chickenpox—are a common presentation of **self-healing LCH**, with spontaneous clearance typically occurring by 2 to 3 months of age.

- **Letterer-Siwe disease (LSD) and Hand-Schüller-Christian disease (HSC):** Cutaneous lesions appear in successive crops, more rapidly in LSD and less so in HSC. On the scalp, lesions may coalesce and mimic **seborrheic dermatitis** or **folliculitis**, often leading to alopecia. In skin folds (retroauricular, axillary, genital), they resemble **intertrigo**.

Lesions may merge into plaques on the medial chest, mid-back, and temporoparietal areas, occasionally becoming **xanthomatous**.

Mucous Membrane Involvement

Mucosal lesions are typically **noduloulcerative** and commonly affect: - Perioral region - Gingiva

In **eosinophilic granuloma (EG)**, perigenital and perianal regions are frequently involved.

Although mucous membrane lesions are absent in HPD, **oral manifestations** may be the initial sign of LCH and include: - Non-specific pain - Aphthous-like ulcers - Gingival bleeding - Candidiasis - Ulceration with tooth loosening - Premature tooth eruption

Nail Changes

Nail abnormalities are associated with a poorer prognosis and include: - Fragile nail plate - Paronychia - Subungual pustules - Nail fold destruction - Onycholysis - Subungual hyperkeratosis - Longitudinal grooving - Pigmented or purpuric striae of the nail bed

These changes can occur in both children and adults.

Cutaneous Findings in Elderly Patients

Skin lesions in older individuals resemble those seen in pediatric cases, supporting the diagnosis of LCH even in adulthood.

Diagnostic Markers for LCH

Immunohistochemical staining of lesional cells typically shows positivity for:

- **S100 protein**
- **CD1a**
- **CD207 (langerin)**

These markers help confirm the presence of Langerhans cells.

Classification of LCH by Extent of Disease

Single-System LCH (Restricted Disease)

- **Skin-limited disease** without involvement of other organs
- **Monostotic bone lesion**, with or without:
 - Diabetes insipidus
 - Adjacent lymph node involvement
 - Rash
- **Polyostotic bone involvement** (multiple bones or >2 lesions in one bone), with or without:
 - Diabetes insipidus
 - Adjacent lymph node involvement
 - Rash

Multisystem LCH (Extensive Disease)

- **Visceral organ involvement** (e.g., liver, spleen, lungs, bone marrow), with or without:
 - Bone lesions
 - Diabetes insipidus
 - Lymph node involvement
 - Rash

Subdivided based on presence of **organ dysfunction**:

- Without dysfunction of lungs, liver, or hematopoietic system
- With dysfunction of lungs, liver, or hematopoietic system (high-risk disease)

Related Physical Findings

Systemic symptoms are common in aggressive forms of LCH and include: - Malaise - Weight loss - Failure to thrive - Nausea - Myalgia - Arthralgia - Fever

Bone Involvement

Bone lesions are the **most frequent manifestation**, occurring in **80% of LCH cases**.

Bone involvement as the sole site of disease is observed in **50–60% of patients**, highlighting the skeletal system as a primary target.